

Chapter 01 Homework — Groups, fields and vector spaces

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Problem 01.3. *Suppose that H is a subgroup of G of index 2. Show that H is a normal subgroup of G .*

We must show that $ghg^{-1} \in H$ for every $g \in G$ and every $h \in H$. Let $h \in H$.

Case 1: $g \in H$. Then $ghg^{-1} \in H$ by closure; no normality is needed to guarantee this.

Case 2: $g \notin H$. Since H has index 2, there are exactly two cosets of H in G , namely H and one other; let gH be that disjoint coset of H , so $g \notin H$. Denote $x = ghg^{-1}$.

Suppose $x \in gH$. Then there exists $h_1 \in H$ such that

$$ghg^{-1} = gh_1 \iff g^{-1} = h^{-1}h_1 \iff g \in H,$$

a contradiction.

So since $x \notin gH$ due to the contradiction, $x \in H$, since H and gH partition G . So $x = ghg^{-1} \in H$ and H is normal. $\square$

Problem 01.5. *Suppose that H is a normal subgroup of G and that K is a normal subgroup of H . Is K necessarily a normal subgroup of G ?*

No. Normality does not pass through; we give a counterexample.

Take $G = S_4$ and let

$$H = \{e, (12)(34), (13)(24), (14)(23)\}.$$

Each non-identity element of H is its own inverse, and the product of any two distinct non-identity elements is the third, so H is a subgroup, and it is abelian. Moreover H consists of *every* element of S_4 of cycle type $2 + 2$, together with e ; since conjugation preserves cycle type, $gHg^{-1} \subseteq H$ for every $g \in S_4$, so $H \trianglelefteq S_4$.

Let $K = \{e, (12)(34)\}$. This is a subgroup, since $(12)(34)$ is its own inverse; any of the three non-identity elements of H would have served equally well, for the same reason. For any $p \in H$ and $k \in K$ we have $pkp^{-1} = k \in K$, because H is abelian, so $K \trianglelefteq H$.

But take $g = (1234) \in G$ and $k = (12)(34) \in K$. Then $g^{-1} = (2143)$ and

$$gkg^{-1} = (1234)(12)(34)(2143) = (1234)(12)(213) = (1234)(13) = (14)(23),$$

and $(14)(23) \notin K$. Hence K is not normal in G . $\square$